

IMAT MOCK EXAM 11: Last Rehearsal

60 questions · 100 minutes · scoring: +1.5 correct, −0.4 wrong, 0 blank · personal target: 42/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Reading Skills, General Knowledge & Logical Reasoning (Q1-9)

1. "Take this medicine at evenly spaced times as prescribed and complete the full course, even if your symptoms improve, because stopping early can let the infection return. If you forget a dose, take it as soon as you remember, but skip the missed dose if your next dose is due soon; never take a double dose. Return any unused tablets to a pharmacy: they must not be kept to treat future illnesses or given to other people."

From the informational leaflet for an antibiotic

Which of the following statements is the correct interpretation of the instructions?

- A)** Stopping the course when symptoms improve is the recommended way to avoid side effects.
- B)** Leftover tablets must not be saved to treat a later illness.
- C)** Two tablets should be taken together whenever a dose is missed.
- D)** This antibiotic should always be taken with food.
- E)** Sharing the medicine with a family member who has similar symptoms is acceptable.

Answer: B

The leaflet states that unused tablets must be returned to a pharmacy and must not be kept to treat future illnesses or given to other people, so the statement about not saving leftover tablets is the correct interpretation. Taking two tablets after a missed dose contradicts the instruction never to take a double dose, and stopping when symptoms improve contradicts completing the full course. Nothing in the passage mentions taking the antibiotic with food, and it explicitly forbids giving the medicine to other people, family members included.

2. "In a large Spanish trial, volunteers at high cardiovascular risk were assigned by lottery to either a Mediterranean-style diet supplemented with extra-virgin olive oil and nuts, or to a control diet advised only to limit fat. Over about five years, the Mediterranean groups suffered roughly 30% fewer heart attacks, strokes and deaths from cardiovascular causes than the control group. The benefit appeared without meaningful weight loss and without a rise in calorie intake. The researchers note that the diet as a whole was tested, not any single ingredient, and that participants lived in a Mediterranean country where these foods are familiar and affordable."

From a newspaper report on nutrition research

Indicate which of the following statements is not correct:

- A) The benefit did not depend on the participants losing weight.
- B) Major cardiovascular events were about 30% lower in the Mediterranean groups.
- C) The researchers evaluated a whole dietary pattern rather than one ingredient.
- D) The trial demonstrated the protective effect of extra-virgin olive oil alone.
- E) Assignment to the Mediterranean or the control diet in the trial was random.

Answer: D

The report says the trial tested the Mediterranean diet as a whole, not any single ingredient, so claiming that it demonstrated a protective effect of extra-virgin olive oil alone is the statement that is not correct. All the other claims restate the passage: assignment was by lottery, so random, cardiovascular events fell by roughly 30%, the benefit appeared without meaningful weight loss, and a whole dietary pattern rather than one component was evaluated. Attributing the effect to a single ingredient goes beyond what a whole-diet trial can show.

3. "With the telescope I have detected four small stars that have escaped the notice of every astronomer before me. They attend a certain planet, arranged in a straight line with it, and their distances from the planet change: at times one hides behind the planet or passes in front of it, and they complete their circuits in unequal periods, the nearest the fastest."

Adapted from Galileo Galilei, Sidereus Nuncius (1610)

What did the author observe?

- A) Four moons orbiting a planet
- B) Spots on the surface of the Sun
- C) Comets passing close to the Sun
- D) Fixed stars that never change position
- E) The rings of Saturn

Answer: A

The four objects attend a planet, line up with it, hide behind it or pass in front of it, and complete circuits at unequal periods with the nearest fastest: bodies orbiting a planet. These are the four large moons of Jupiter, Galileo's discovery of 1610. Sunspots sit on the Sun rather than orbiting it, comets orbit the Sun, genuinely fixed stars show no such motion, and the rings of Saturn are not four separate bodies that repeatedly disappear behind a planet in a line with it.

4. "An antibiotic kills susceptible bacteria but spares any cells that already carry resistance genes, for example a gene encoding an enzyme that destroys the drug. The surviving cells multiply, and resistance genes can also pass between bacteria on small circles of DNA. In environments where antibiotics are used often, such as hospitals, resistant strains therefore make up a growing share of the bacterial population over time."

From a microbiology textbook

Only one of the following statements is coherent with the excerpt:

- A) Antibiotics create resistance genes inside susceptible bacteria.
- B) Susceptible and resistant cells are killed by an antibiotic to the same degree.
- C) Bacteria deliberately mutate when an antibiotic is present so that they can survive it.
- D) Resistance genes can never move from one bacterium to another.
- E) Frequent use of antibiotics favours the spread of resistant strains.

Answer: E

Because the antibiotic kills susceptible cells and spares those already carrying resistance genes, frequent use, as in hospitals, selects for the resistant strains, which then multiply and share genes on small DNA circles; so the statement that frequent antibiotic use favours the spread of resistant strains is coherent with the excerpt. Selection needs no purpose, so bacteria do not deliberately mutate to survive, and the drug does not create resistance genes, it only favours cells that already have them. The excerpt says the genes do pass between bacteria, and susceptible cells are by definition killed more than resistant ones.

5. Consider this statement: All squares are rectangles.

Which of the following statements is deducible from the given statement?

- A) All rectangles are squares.
- B) Some rectangles are not squares.
- C) If a figure is not a rectangle, then it is not a square.
- D) Every four-sided figure is a rectangle.
- E) Squares and rectangles share no geometric properties.

Answer: C

Restate the given statement: every square is a rectangle, so the squares form a subclass of the rectangles.

Deduction 1: a figure that is not a rectangle lies outside the whole class of rectangles.

Deduction 2: since every square lies inside that class, a figure outside it cannot be a square.

Conclusion: if a figure is not a rectangle, then it is not a square; this is the contrapositive, which always follows from a universal statement.

The tempting claim that some rectangles are not squares is true in ordinary geometry but not deducible here, because the premise would hold equally well if every rectangle were a square, and reversing the claim into all rectangles are squares is the invalid converse.

6. From the given statements:

If it rains this afternoon, the tennis match will be moved indoors.

If the match is moved indoors, it will start at 2 pm.

The match did not start at 2 pm.

Which conclusion can be deduced?

- A)** The match started late.
- B)** It did not rain this afternoon.
- C)** The match was cancelled.
- D)** The match was moved indoors but started at a different time.
- E)** It rained this afternoon.

Answer: B

Restate the chain: if it rains, the match is moved indoors; if it is moved indoors, it starts at 2 pm; and it is given that the match did not start at 2 pm.

Deduction 1: moved indoors leads to a 2 pm start, and there was no 2 pm start, so by modus tollens the match was not moved indoors.

Deduction 2: rain leads to an indoor move, and there was no indoor move, so by modus tollens again it did not rain this afternoon.

Conclusion: it did not rain this afternoon.

The tempting claim that the match started late fails because the premises name no other starting time at all, and cancellation or an indoor start at a different hour would add facts the premises never supply.

7. Amara, Bruno and Chiara live on three different floors of a small building, one person per floor. Bruno lives on the third floor, Amara lives directly above Bruno, and Chiara lives directly above Amara. On which floor does Chiara live?

- A)** On the sixth floor.
- B)** On the fourth floor.
- C)** On the third floor.
- D)** On the second floor.
- E)** On the fifth floor.

Answer: E

Restate: one person per floor; Bruno is on the third floor; Amara lives directly above Bruno; Chiara lives directly above Amara.

Deduction 1: Bruno is fixed on floor 3.

Deduction 2: directly above floor 3 is floor 4, so Amara lives on floor 4.

Deduction 3: directly above floor 4 is floor 5, so Chiara lives on floor 5.

Conclusion: Chiara lives on the fifth floor.

The tempting fourth floor is already occupied by Amara, and every floor below the fifth would contradict the two directly-above conditions that force Chiara two storeys above Bruno.

8. Three students, Sara, Tom and Leo, each study exactly one foreign language among French, German and Spanish. Sara studies neither French nor German, and Leo studies German. Which of the following is correct?

- A) Tom studies French.
- B) Tom studies German.
- C) Tom studies Spanish.
- D) Sara studies French.
- E) Leo studies Spanish.

Answer: A

Restate: three students take exactly one language each among French, German and Spanish; Sara takes neither French nor German; Leo takes German.

Deduction 1: Sara's two exclusions leave only Spanish, so Sara studies Spanish.

Deduction 2: Leo is already fixed on German.

Deduction 3: French and German are taken and Sara holds Spanish, so the only language left for Tom is French.

Conclusion: Tom studies French.

Any claim giving Tom German or Spanish fails because those languages are already assigned to Leo and Sara, and giving Sara French contradicts an explicit exclusion.

9. What is the angle between the hour hand and the minute hand of an analogue clock at exactly 9:30?

- A) 75°
- B) 90°
- C) 105°
- D) 120°
- E) 180°

Answer: C

Given: the time 9:30; each hour mark on a clock face spans 30° and the hour hand moves 30° per hour.

Minute hand: at 30 minutes it points exactly at the 6, so its angle clockwise from 12 is $6 \times 30^\circ = 180^\circ$.

Hour hand: it starts at the 9, at $9 \times 30^\circ = 270^\circ$, and drifts $0.5 \text{ h} \times 30^\circ/\text{h} = 15^\circ$ during the half hour, giving $270^\circ + 15^\circ = 285^\circ$.

Difference: $285^\circ - 180^\circ = 105^\circ$

Result: the angle between the hands is 105°.

The tempting 90° freezes the hour hand exactly on the 9 and ignores its 15° half-hour drift.

Section B - Biology (Q10-32)

10. The small amount of ATP that glycolysis itself produces is formed by:

- A) oxidation of pyruvate to acetyl-CoA.
- B) hydrolysis of the NADH produced in the pathway.
- C) chemiosmosis at the inner mitochondrial membrane.
- D) direct transfer of a phosphate group from a phosphorylated intermediate to ADP.
- E) the electron transport chain.

Answer: D

Glycolysis makes its small ATP yield by substrate-level phosphorylation: phosphorylated intermediates such as 1,3-bisphosphoglycerate and phosphoenolpyruvate transfer a phosphate group straight to ADP. Chemiosmosis at the inner mitochondrial membrane and the electron transport chain belong to the later stage of oxidative phosphorylation, oxidising pyruvate to acetyl-CoA yields no ATP, and the NADH made in glycolysis is cashed in at the chain, not by hydrolysis.

11. The reaction that converts pyruvate into acetyl-CoA before entry into the Krebs cycle takes place in:
- A) the cytosol.
 - B) the intermembrane space.
 - C) the Golgi apparatus.
 - D) the inner mitochondrial membrane.
 - E) the mitochondrial matrix.

Answer: E

Pyruvate made by cytosolic glycolysis is transported across the inner mitochondrial membrane into the matrix, where the pyruvate dehydrogenase complex oxidatively decarboxylates it to acetyl-CoA. The link reaction therefore occurs in the mitochondrial matrix, not in the cytosol where glycolysis runs, not in the intermembrane space where protons accumulate, and not at the inner membrane where the electron transport chain sits. The Golgi apparatus plays no part in respiration.

12. During the complete oxidation of one glucose molecule to CO₂ and water, carbon dioxide is released during:
- A) glycolysis and oxidative phosphorylation.
 - B) glycolysis and the Krebs cycle.
 - C) the link reaction and the Krebs cycle.
 - D) the Krebs cycle and oxidative phosphorylation.
 - E) glycolysis and the link reaction.

Answer: C

Step 1: glycolysis splits glucose into two pyruvates and releases no carbon dioxide. Step 2: each pyruvate loses one carbon as CO₂ on conversion to acetyl-CoA in the link reaction, 2 CO₂ in total. Step 3: two turns of the Krebs cycle, one per acetyl group, release the remaining 4 CO₂. Oxidative phosphorylation merely passes electrons to oxygen and forms water, so carbon dioxide appears only in the link reaction and the Krebs cycle.

13. A substance makes the inner mitochondrial membrane leaky to protons while leaving the electron transport complexes untouched. The most direct consequences are that:
- A) electron transport stops while ATP synthesis continues.
 - B) electron transport accelerates while ATP synthesis falls.
 - C) ATP synthesis increases while electron transport slows.
 - D) the proton gradient grows and ATP synthesis accelerates.
 - E) electron transport and ATP synthesis both stop.

Answer: B

A leak that lets protons cross the inner membrane without passing through ATP synthase discharges the proton gradient, so the driving force for ATP synthesis collapses and ATP output falls. With the gradient no longer holding the electron chain back, electron transport speeds up rather than stopping. Both processes stopping would require poisoning the complexes themselves, which the stem explicitly rules out; this is precisely how uncouplers behave.

14. In the mitochondrial respiratory chain, electrons are carried from complex III to complex IV by:
- A) cytochrome c.
 - B) ubiquinone.
 - C) oxygen.
 - D) NAD⁺.
 - E) ATP synthase.

Answer: A

In the chain, complexes I and II pass electrons to ubiquinone, which delivers them to complex III; from complex III the electrons travel to complex IV on cytochrome c, a small mobile protein on the outer face of the inner membrane, and oxygen finally accepts them at complex IV. Ubiquinone therefore works before complex III, oxygen is the terminal acceptor rather than a shuttle, NAD⁺ is not a respiratory-chain carrier, and ATP synthase is powered by protons rather than carrying electrons.

15. In haemoglobin, each oxygen molecule binds to:

- A)** a molecule of bicarbonate inside the cell.
- B)** the membrane of the red blood cell.
- C)** a terminal amino group of a globin chain.
- D)** the iron ion at the centre of a heme group.
- E)** a hydrophobic pocket between two subunits.

Answer: D

Each of the four globin subunits of haemoglobin carries a heme group, and the iron ion at its centre binds one oxygen molecule reversibly, giving four O₂ per haemoglobin molecule. Carbon dioxide instead travels mainly as bicarbonate and partly on the terminal amino groups of the globin chains, and neither the red cell membrane, nor a hydrophobic pocket between subunits, nor a bicarbonate ion is the oxygen-binding site.

16. The largest fraction of the carbon dioxide carried in venous blood is transported:

- A)** as bicarbonate ions in the plasma.
- B)** inside the organelles of red blood cells.
- C)** as carbamino compounds on plasma proteins.
- D)** dissolved directly in the plasma.
- E)** bound to the iron atoms of haemoglobin.

Answer: A

Carbonic anhydrase inside red blood cells converts most incoming CO₂ to carbonic acid, which dissociates to bicarbonate; the bicarbonate then leaves the cell for the plasma in exchange for chloride, and about 70% of the CO₂ travels this way, by far the largest share. Only around a fifth binds to haemoglobin as carbamino compounds and a few percent dissolves directly. The heme iron binds oxygen, not carbon dioxide, and mature red blood cells contain no organelles at all.

17. Compared with fast-twitch fibres, slow-twitch (type I) muscle fibres:

- A)** contain fewer mitochondria and fatigue rapidly.
- B)** rely mainly on anaerobic glycolysis for their ATP.
- C)** contain more mitochondria and are more resistant to fatigue.
- D)** contract more rapidly but for shorter periods.
- E)** store no myoglobin at all.

Answer: C

Slow-twitch type I fibres are endurance fibres packed with mitochondria and myoglobin and richly supplied with capillaries, so their oxidative metabolism lets them contract for long periods without fatiguing. Fast-twitch fibres are the ones with fewer mitochondria that rely mainly on anaerobic glycolysis and tire rapidly. Slow fibres are far from lacking myoglobin; their high myoglobin content is what makes them red.

18. The thin layer of cells lining the inner surface of the heart chambers and of all blood vessels is called the:

- A)** mesothelium.
- B)** endothelium.
- C)** periosteum.
- D)** collagen sheath.
- E)** perichondrium.

Answer: B

The simple squamous epithelium that lines the heart chambers and every blood vessel is called the endothelium. Mesothelium instead lines the closed body cavities such as the pleural and pericardial spaces, perichondrium wraps cartilage, periosteum wraps bone, and a collagen sheath is extracellular connective tissue rather than a cellular lining.

19. The activity of an enzyme rises with temperature up to an optimum and then falls steeply at higher temperatures because the enzyme:

- A)** switches to catalysing a different reaction.
- B)** has run out of substrate.
- C)** is broken down into its component fatty acids.
- D)** starts to synthesise an inhibitor of itself.
- E)** begins to denature, losing its functional three-dimensional shape.

Answer: E

Above the optimum temperature, extra heat breaks the weak hydrogen and ionic interactions that hold the enzyme's three-dimensional fold, so the protein denatures and the active site loses its shape, and activity falls steeply. Substrate supply does not change with temperature, enzymes are proteins that break down to amino acids rather than fatty acids, and no enzyme switches reaction or manufactures its own inhibitor on heating.

20. Inhibition by a non-competitive inhibitor differs from competitive inhibition in that it:

- A)** affects only the apparent affinity of the enzyme for its substrate.
- B)** is reversed completely by adding excess substrate.
- C)** disappears when the reaction mixture is cooled.
- D)** lowers the maximum rate of the reaction and is not overcome by excess substrate.
- E)** involves the inhibitor binding at the active site.

Answer: D

A non-competitive inhibitor binds at an allosteric site away from the active site and inactivates those enzyme molecules completely, so the maximum reaction rate falls and no excess of substrate can restore it. Being fully reversed by excess substrate, and changing only the apparent affinity for substrate, are the defining behaviours of competitive inhibition, which occupies the active site itself. Temperature has nothing specific to do with the distinction.

21. Between meals, bile is stored and concentrated in the:

- A)** liver.
- B)** stomach.
- C)** small intestine.
- D)** pancreas.
- E)** gall bladder.

Answer: E

The liver produces bile continuously, but between meals the bile is diverted into the gall bladder, which stores it and concentrates it by reabsorbing water until a fatty meal triggers its contraction into the duodenum. The pancreas produces digestive enzymes and bicarbonate but no bile, and the small intestine is where bile acts, not where it is stored between meals.

22. Soon after a meal rich in carbohydrates, the liver responds to the prevailing hormonal signal by:

- A)** converting glucose into glycogen for storage.
- B)** converting excess glucose into urea.
- C)** breaking glycogen down into free glucose.
- D)** increasing the loss of glucose in the bile.
- E)** secreting glucagon into the blood.

Answer: A

A carbohydrate-rich meal raises blood glucose, which stimulates insulin secretion, and insulin makes the liver polymerise glucose into glycogen for storage, glycogenesis. Breaking glycogen down to free glucose is the opposite, the fasting response driven by glucagon, whose secretion falls at this time; glucagon is released by the pancreas, not the liver; and glucose contains no nitrogen, so it cannot be turned into urea.

23. The osmotic gradient of the renal medulla, which allows the production of concentrated urine, is created mainly by active salt transport in the:

- A)** proximal convoluted tubule.
- B)** distal convoluted tubule.
- C)** loop of Henle.
- D)** glomerulus.
- E)** renal artery.

Answer: C

The thick ascending limb of the loop of Henle actively pumps sodium and chloride into the medullary tissue while remaining impermeable to water, creating the osmotic gradient that the collecting duct, under ADH, later uses to reabsorb water and produce concentrated urine. The glomerulus only filters blood, the proximal tubule reabsorbs glucose and most solutes without building the gradient, the distal tubule fine-tunes ion balance, and a blood vessel cannot generate a tissue gradient.

24. The cells whose beating cilia sweep mucus and trapped dust upward out of the airways are:

- A) alveolar macrophages.
- B) ciliated epithelial cells.
- C) chondrocytes.
- D) type I alveolar cells.
- E) goblet cells.

Answer: B

Ciliated epithelial cells line the conducting airways, and their coordinated ciliary beating transports the overlying mucus blanket with its trapped dust up to the pharynx, where it is swallowed, the mucociliary escalator. Goblet cells only secrete the mucus, alveolar macrophages patrol the alveoli deep in the lung, type I alveolar cells form the gas-exchange wall, and chondrocytes build the cartilage rings.

25. Iodine is required in the human diet mainly for the synthesis of:

- A) cortisol.
- B) adrenaline.
- C) antidiuretic hormone.
- D) thyroxine.
- E) insulin.

Answer: D

The thyroid is the only tissue that concentrates dietary iodide, which it incorporates into thyroxine (T4) and triiodothyronine (T3); chronic iodine shortage therefore enlarges the gland into a goitre. Insulin, adrenaline, antidiuretic hormone and cortisol contain no iodine at all, so none of them can be the answer.

26. The α -helix and the β -pleated sheet are features of which level of protein structure?

- A) Primary structure
- B) Secondary structure
- C) The prosthetic group level
- D) Tertiary structure
- E) Quaternary structure

Answer: B

The alpha-helix and the beta-pleated sheet are regular local folds of the polypeptide backbone stabilised by hydrogen bonds between backbone groups, which is exactly the definition of secondary structure. Primary structure is the amino acid sequence, tertiary structure is the complete three-dimensional fold of one chain, quaternary structure is the assembly of several chains, and no prosthetic-group level exists in the standard scheme.

27. According to the fluid mosaic model of the plasma membrane, the proteins of the membrane:

- A) can drift laterally within the lipid bilayer.
- B) are composed of carbohydrate.
- C) form a solid continuous coat over the bilayer.
- D) occur only on the outer surface of the membrane.
- E) are each fixed rigidly in a single position.

Answer: A

In the fluid mosaic model the bilayer is a two-dimensional fluid in which both lipids and integral proteins can drift laterally, classically shown when fluorescently labelled proteins of a fused mouse and human cell intermixed across the hybrid membrane. The proteins are not rigidly fixed in place, they span the membrane or sit on both faces rather than only the outer surface, they are proteins rather than carbohydrate, and the membrane is a mosaic, not a solid continuous coat.

28. The digestive enzymes inside lysosomes have an optimum pH of about:

- A)** 2. **B)** 5. **C)** 7.4. **D)** 9. **E)** 12.

Answer: B

Lysosomal hydrolases have their optimum near pH 5, maintained by proton pumps that acidify the organelle's interior. A pH near 2 is typical of the stomach lumen, 7.4 of blood, and 9 or 12 are far too alkaline. The mildly acidic optimum is also a safety feature: any hydrolase escaping into the cytosol at about pH 7.2 is largely inactivated.

29. Catalase, the enzyme that decomposes hydrogen peroxide into water and oxygen, is concentrated in which organelle?

- A)** The Golgi apparatus
B) The mitochondrion
C) The nucleolus
D) The lysosome
E) The peroxisome

Answer: E

Catalase is the signature enzyme of the peroxisome, an organelle in which oxidase reactions produce hydrogen peroxide and catalase promptly decomposes it to water and oxygen. Lysosomes instead digest macromolecules at acidic pH, mitochondria make ATP, the Golgi apparatus modifies and packages proteins, and the nucleolus assembles ribosomal subunits.

30. For a cross between two heterozygotes, $Aa \times Aa$, the expected ratio of genotypes $AA : Aa : aa$ among the offspring is:

- A)** 1:1:1 **B)** 1:2:1 **C)** 2:1:1 **D)** 3:1 **E)** 9:3:3:1

Answer: B

Given: both parents are heterozygous, $Aa \times Aa$.

Step 1: each parent's gametes carry A or a, 1/2 each.

Step 2: Punnett square outcomes: AA with probability 1/4, Aa with probability 1/2 (two ways: A then a, or a then A), aa with probability 1/4.

Step 3: genotypes therefore appear in the ratio $AA : Aa : aa = 1 : 2 : 1$.

Step 4: the question asks for the genotype ratio, not the phenotype ratio.

Result: 1 : 2 : 1.

Classic slip: 3 : 1 is the phenotype ratio, because AA and Aa look alike; 9 : 3 : 3 : 1 belongs to a dihybrid cross.

31. Red-green colour blindness is caused by a recessive allele carried on the X chromosome. A man with colour blindness and a woman with normal vision whose father was colour blind have children. Which statement is correct?

- A)** All of the children will have normal vision.
- B)** All of the daughters will be colour blind.
- C)** On average, half of the sons and half of the daughters will be colour blind.
- D)** All of the sons will be colour blind.
- E)** The daughters cannot have received the allele from their father.

Answer: C

Given: colour blindness is caused by a recessive allele on the X chromosome; the man is affected, so his only X chromosome carries the allele; the woman has normal vision but her father was colour blind, so her second X must carry the allele and she is a carrier.

Step 1: half the man's sperm carry his affected X, half carry Y.

Step 2: half the woman's eggs carry her normal X, half carry her affected X.

Step 3: combining equally: affected X + normal X = carrier daughter; affected X + affected X = colour-blind daughter; Y + normal X = normal son; Y + affected X = colour-blind son, each with probability 1/4.

Step 4: so half the sons and half the daughters are colour blind.

Result: on average half of the sons and half of the daughters are colour blind.

Classic slip: every daughter does receive the affected X from the father, but half of the daughters also receive a normal X from the mother and so are only carriers.

32. During elongation of translation, the new peptide bond forms between the growing polypeptide chain and:

- A)** the anticodon of the following tRNA.
- B)** the ribosomal RNA of the small subunit.
- C)** the tRNA about to leave through the E site.
- D)** the amino acid carried by the tRNA that has just entered the A site.
- E)** the codon currently being read in the mRNA.

Answer: D

During elongation the peptidyl transferase activity of the large ribosomal subunit transfers the growing polypeptide from the tRNA in the P site onto the amino acid held by the tRNA that has just entered the A site, so each new peptide bond forms at that newly arrived aminoacyl-tRNA. Codons and anticodons carry information but no amino acid, the E-site tRNA leaves only after it has already surrendered its amino acid, and ribosomal RNA is the catalytic scaffold rather than a bond partner.

Section C - Chemistry (Q33-47)

33. What is the pH of a 0.001 mol/dm^3 aqueous solution of sodium hydroxide, a strong base? ($K_w = 1.0 \times 10^{-14}$ at 25°C)

- A)** 3.0 **B)** 7.0 **C)** 10.0 **D)** 11.0 **E)** 13.0

Answer: D

Given: $c(\text{NaOH}) = 0.001 \text{ mol/dm}^3 = 1.0 \times 10^{-3} \text{ mol/dm}^3$, a strong base; $K_w = 1.0 \times 10^{-14}$ at 25°C

Relationship: a strong base dissociates completely, so $[\text{OH}^-]$ equals its concentration; $\text{pOH} = -\log[\text{OH}^-]$ and $\text{pH} + \text{pOH} = 14$.

Substitute: $[\text{OH}^-] = 1.0 \times 10^{-3} \text{ mol/dm}^3$, so $\text{pOH} = -\log(1.0 \times 10^{-3})$

Step: $\text{pOH} = 3.0$

Step: $\text{pH} = 14.0 - 3.0 = 11.0$

Result: the pH of the solution is 11.0.

The tempting 3.0 is the pOH itself misreported as the pH, and 13.0 would belong to a ten times more concentrated 0.1 mol/dm^3 solution.

34. At 25°C , pure water has $K_w = 1.0 \times 10^{-14}$ and pH 7. At 80°C , K_w rises to 4.0×10^{-13} . In pure water at 80°C , the concentration of H^+ and the nature of the water are:

- A)** $[\text{H}^+] = 1.0 \times 10^{-7} \text{ mol/dm}^3$ and the water is neutral.
B) $[\text{H}^+] = 6.3 \times 10^{-7} \text{ mol/dm}^3$ and the water is acidic.
C) $[\text{H}^+] = 6.3 \times 10^{-7} \text{ mol/dm}^3$ and the water is neutral.
D) $[\text{H}^+] = 2.0 \times 10^{-13} \text{ mol/dm}^3$ and the water is basic.
E) $[\text{H}^+] = 4.0 \times 10^{-13} \text{ mol/dm}^3$ and the water is acidic.

Answer: C

Given: pure water at 80°C with $K_w = 4.0 \times 10^{-13}$

Relationship: pure water always produces H^+ and OH^- in equal numbers, so $[\text{H}^+] = [\text{OH}^-] = \sqrt{K_w}$; neutrality means this equality of the two ions, not a pH of exactly 7.

Substitute: $[\text{H}^+] = \sqrt{(4.0 \times 10^{-13}) \text{ mol/dm}^3} = \sqrt{(40 \times 10^{-14}) \text{ mol/dm}^3}$

Step: $\sqrt{40} \approx 6.3$, so $[\text{H}^+] \approx 6.3 \times 10^{-7} \text{ mol/dm}^3$, a pH of about 6.2.

Result: $[\text{H}^+] \approx 6.3 \times 10^{-7} \text{ mol/dm}^3$ and the water is neutral.

The tempting $1.0 \times 10^{-7} \text{ mol/dm}^3$ recycles the 25°C value and ignores the new temperature, and setting $[\text{H}^+]$ equal to K_w itself confuses the ion product with an ion concentration.

35. Which of the following substances has a giant covalent (network covalent) structure?

- A)** Diamond
B) Sodium chloride
C) Methane
D) Ice
E) Copper

Answer: A

Diamond is one continuous lattice in which every carbon atom is covalently bonded to four neighbours throughout the crystal, the defining giant covalent network. Sodium chloride is a giant ionic lattice, copper is a giant metallic lattice, and ice and methane consist of discrete molecules: ice molecules cling to each other only by hydrogen bonds and methane molecules only by weak dispersion forces.

36. Water ($M = 18 \text{ g/mol}$) boils at 100°C , while methane ($M = 16 \text{ g/mol}$) boils at -162°C . The main reason for the much higher boiling point of water is:

- A)** ionic bonds between water molecules.
- B)** the larger number of covalent bonds inside each water molecule.
- C)** the slightly greater molecular mass of water.
- D)** covalent bonds linking the atoms of different molecules.
- E)** hydrogen bonds between water molecules.

Answer: E

Each water molecule can both donate and accept hydrogen bonds through its O-H groups and oxygen lone pairs, and this strong intermolecular attraction must be overcome to vaporise the liquid, hence the boiling point of 100°C . There are no ionic bonds between water molecules, and the covalent bonds inside each molecule are not broken on boiling. A mass difference of just 2 g/mol cannot explain a gap of about 260 degrees, since methane's mass gives it only weak dispersion forces.

37. What is the oxidation number of phosphorus in H_3PO_4 ?

- A)** +1 **B)** +3 **C)** +5 **D)** +6 **E)** +7

Answer: C

Given: H_3PO_4 , a neutral molecule; standard oxidation numbers $\text{H} = +1$ and $\text{O} = -2$.

Relationship: oxidation numbers sum to zero over a neutral compound; write the unknown value for phosphorus as x .

Substitute: $3 \times (+1) + x + 4 \times (-2) = 0$

Step: $3 + x - 8 = 0$

Step: $x - 5 = 0$

Result: $x = +5$, so phosphorus is in oxidation state +5

The tempting +3 belongs to phosphorous acid, H_3PO_3 , which carries one oxygen fewer than phosphoric acid.

38. In the reaction $2\text{FeCl}_2 + \text{Cl}_2 \rightarrow 2\text{FeCl}_3$, which species is the oxidising agent?

- A)** Fe^{2+} , because it combines with chlorine.
- B)** Fe^{3+} , because it is the product of the reaction.
- C)** Cl^- , because it carries a negative charge.
- D)** Neither species, because no electrons are transferred.
- E)** Cl_2 , because it accepts electrons from the iron and is itself reduced.

Answer: E

Restate: in $2\text{FeCl}_2 + \text{Cl}_2 \rightarrow 2\text{FeCl}_3$ the oxidising agent is the species that gains electrons and is itself reduced.

Deduction 1: iron rises from +2 in FeCl_2 to +3 in FeCl_3 , so each iron ion loses one electron and is oxidised.

Deduction 2: chlorine falls from 0 in Cl_2 to -1 in the chloride ions, so each chlorine atom gains one electron and is reduced.

Conclusion: Cl_2 accepts the electrons given up by the iron, so Cl_2 is the oxidising agent and the iron(II) is the reducing agent.

The tempting identification of the iron(II) fails because it donates electrons rather than accepting them, and a negative charge alone never defines the role: the chloride in the products is the already-reduced species, not the electron acceptor.

39. How many open-chain alcohols have the molecular formula $C_4H_{10}O$?

- A) 2 B) 3 C) 4 D) 5 E) 6

Answer: C

Given: molecular formula $C_4H_{10}O$, open-chain alcohols only, so one OH group on a four-carbon skeleton (ethers are excluded by the question).

Step 1: four carbons admit exactly two skeletons, straight-chain butane and branched 2-methylpropane.

Step 2: on butane, positions 1 and 4 are equivalent and positions 2 and 3 are equivalent, so the OH gives butan-1-ol and butan-2-ol, 2 isomers.

Step 3: on 2-methylpropane, the OH on any of the three equivalent outer carbons gives 2-methylpropan-1-ol and on the single central carbon gives 2-methylpropan-2-ol, 2 isomers.

Step: $2 + 2 = 4$ isomers

Result: 4 open-chain alcohols.

The tempting count of 2 forgets the branched skeleton, and adding the open-chain ethers would overcount, since the question restricts the answer to alcohols.

40. The bond formed between the amino group of one amino acid and the carboxyl group of another is called:

- A) a phosphodiester bond.
B) an ester bond.
C) a glycosidic bond.
D) a peptide bond.
E) a hydrogen bond.

Answer: D

The condensation reaction between the carboxyl group of one amino acid and the amino group of another releases water and forms the $-CO-NH-$ linkage called a peptide bond. Glycosidic bonds join monosaccharides, ester bonds join glycerol to fatty acids, phosphodiester bonds join nucleotides in nucleic acids, and a hydrogen bond is a weak non-covalent attraction rather than a true covalent linkage.

41. What amount, in moles, is present in 8.8 g of carbon dioxide? ($M(CO_2) = 44 \text{ g/mol}$)

- A) 0.1 mol B) 0.2 mol C) 0.44 mol D) 2.0 mol E) 4.4 mol

Answer: B

Given: mass of $CO_2 = 8.8 \text{ g}$; $M(CO_2) = 44 \text{ g/mol}$

Relationship: amount of substance $n = \text{mass} \div \text{molar mass}$.

Substitute: $n = 8.8 \text{ g} \div 44 \text{ g/mol}$

Step: $8.8 \div 44 = 0.2$, so $n = 0.2 \text{ mol}$

Result: 0.2 mol of carbon dioxide.

The tempting 0.44 mol comes from dividing 8.8 by 20 instead of 44, and values above 1 mol come from multiplying the mass by the molar mass instead of dividing.

42. A sample of gas occupies 4.0 dm^3 at 300 K and 100 kPa . If the sample is heated to 600 K at constant pressure, its volume becomes:

- A)** 2.0 dm^3 **B)** 4.0 dm^3 **C)** 6.0 dm^3 **D)** 8.0 dm^3 **E)** 12.0 dm^3

Answer: D

Given: $V_1 = 4.0 \text{ dm}^3$ at $T_1 = 300 \text{ K}$; heated at constant pressure to $T_2 = 600 \text{ K}$

Relationship: Charles's law, $V_1 \div T_1 = V_2 \div T_2$, so $V_2 = V_1 \times (T_2 \div T_1)$

Substitute: $V_2 = 4.0 \text{ dm}^3 \times (600 \text{ K} \div 300 \text{ K})$

Step: $600 \div 300 = 2$, so $V_2 = 4.0 \text{ dm}^3 \times 2 = 8.0 \text{ dm}^3$

Result: the volume becomes 8.0 dm^3 .

The tempting 2.0 dm^3 halves the volume on heating, reversing the direct proportionality; a gas shrinks only when its absolute temperature falls.

43. Going down group 1 of the periodic table, from lithium towards francium, the reactivity of the metals with water:

- A)** increases, because the outer electron is further from the nucleus and is lost more easily.
B) remains constant, because every group 1 metal has one outer electron.
C) decreases, because the outer electron is held more tightly.
D) decreases, because the atoms become smaller.
E) first increases and then decreases.

Answer: A

Going down group 1 each element adds a shell, so the atoms grow larger and the single outer electron sits further from the nucleus with more shielding between them; the attraction holding it weakens, it is lost more easily, and reactivity with water increases from lithium towards caesium. The atoms do not shrink down the group, and having one outer electron in common does not make reactivity constant, because what matters is how easily that electron escapes.

44. The electron configuration of a neutral atom of sulfur ($Z = 16$) is:

- A)** $1s^2 2s^2 2p^6 3s^2 3p^4$
B) $1s^2 2s^2 2p^6 3s^2 3p^6$
C) $1s^2 2s^2 2p^6 3s^1 3p^5$
D) $1s^2 2s^2 2p^5 3s^2 3p^5$
E) $1s^2 2s^2 2p^4 3s^2 3p^6$

Answer: A

Given: neutral sulfur, $Z = 16$, so 16 electrons to place in the order $1s, 2s, 2p, 3s, 3p$.

Step 1: $1s^2$ takes 2 electrons, running total 2.

Step 2: $2s^2$ takes 2, running total 4.

Step 3: $2p^6$ takes 6, running total 10.

Step 4: $3s^2$ takes 2, running total 12.

Step 5: the remaining $16 - 12 = 4$ electrons fill $3p$, giving $3p^4$.

Result: $1s^2 2s^2 2p^6 3s^2 3p^4$.

The tempting configuration ending in $3p^6$ holds 18 electrons in total, which is argon rather than sulfur, and the other variants keep 16 only by illegally emptying $2s$ or $2p$.

45. For the equilibrium $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, the equilibrium constant K is given by:

- A) $K = 2[\text{SO}_3] / (2[\text{SO}_2] + [\text{O}_2])$
- B) $K = ([\text{SO}_2]^2[\text{O}_2]) / [\text{SO}_3]^2$
- C) $K = [\text{SO}_3] / ([\text{SO}_2]^2[\text{O}_2])$
- D) $K = [\text{SO}_3]^2 / ([\text{SO}_2][\text{O}_2])$
- E) $K = [\text{SO}_3]^2 / ([\text{SO}_2]^2[\text{O}_2])$

Answer: E

Given: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ with coefficients 2, 1 and 2.

Relationship: the equilibrium constant equals the product concentrations divided by the reactant concentrations, each raised to the power of its stoichiometric coefficient.

Substitute: numerator $[\text{SO}_3]$ to the power 2; denominator $[\text{SO}_2]$ to the power 2 multiplied by $[\text{O}_2]$ to the power 1.

$$\text{Result: } K = [\text{SO}_3]^2 \div ([\text{SO}_2]^2 \times [\text{O}_2])$$

The tempting version with plain $[\text{SO}_2]$ in the denominator forgets that the coefficient 2 becomes an exponent on the SO_2 concentration, and the inverted fraction swaps products and reactants.

46. When the temperature of a reaction mixture is raised, the rate of the reaction increases. Which property of the reaction remains unchanged?

- A) The rate at which products form.
- B) The number of reactant molecules able to react.
- C) The activation energy of the reaction.
- D) The frequency of collisions between reactant molecules.
- E) The fraction of molecular collisions energetic enough to react.

Answer: C

Heating reshapes the Maxwell-Boltzmann energy distribution so that a larger fraction of collisions exceeds the activation energy, and collisions also occur somewhat more frequently; both effects raise the rate. The activation energy itself is fixed by the reaction pathway in the energy profile and does not depend on temperature, so it is the property that stays unchanged unless a catalyst alters the mechanism. The rate is by definition what increases, and the fraction of sufficiently energetic collisions rises, so neither can be the unchanged quantity.

47. The compound formed between the aluminium ion, Al^{3+} , and the sulfate ion, SO_4^{2-} , has the formula:

- A) Al_2SO_4
- B) $\text{Al}_2(\text{SO}_4)_3$
- C) $\text{Al}(\text{SO}_4)_3$
- D) $\text{Al}_3(\text{SO}_4)_2$
- E) AlSO_4

Answer: B

Given: aluminium ion Al^{3+} with charge +3; sulfate ion SO_4^{2-} with charge -2.

Relationship: a neutral compound needs equal total positive and negative charge, so take the least common multiple of the charge magnitudes.

Step: $\text{LCM}(3, 2) = 6$, so the compound needs +6 from aluminium and -6 from sulfate.

Step: $6 \div 3 = 2$ aluminium ions and $6 \div 2 = 3$ sulfate ions.

Result: $\text{Al}_2(\text{SO}_4)_3$, with parentheses because the polyatomic sulfate group appears three times.

The tempting AlSO_4 leaves a net charge of +1, and swapping the subscripts to $\text{Al}_3(\text{SO}_4)_2$ gives +9 against -4, so neither is neutral.

Section D - Mathematics (Q48-54)

48. Solve the inequality $1 - 2x \geq 5$.

- A)** $x \leq -2$ **B)** $x \geq 2$ **C)** $x \geq -2$ **D)** $x \leq -1$ **E)** $x \leq 2$

Answer: A

Given: $1 - 2x \geq 5$.

Step 1: subtract 1 from both sides, giving $-2x \geq 5 - 1 = 4$.

Step 2: divide both sides by -2 ; dividing by a negative number reverses the inequality sign.

Step 3: $x \leq 4 \div (-2) = -2$

Result: $x \leq -2$.

The tempting $x \geq -2$ forgets the sign reversal, and the answers ending at $+2$ come from losing the minus sign on the constant or on the coefficient.

49. The set of real solutions of the inequality $(x - 2)^2 \geq 0$ is:

- A)** $x \geq 2$
B) only $x = 2$
C) the empty set
D) the set of all real numbers
E) $x \leq 2$

Answer: D

Restate: find every real x satisfying $(x - 2)^2 \geq 0$.

Deduction 1: the square of a real number is never negative, so $(x - 2)^2 \geq 0$ holds automatically for every x .

Deduction 2: equality holds only at $x = 2$, but the inequality is satisfied by that value and by every other real number as well.

Conclusion: the solution set is all real numbers.

The tempting restriction to only $x = 2$ solves the stricter equation $(x - 2)^2 = 0$ instead of the given inequality, and the empty set would require a condition no real number can meet, such as a square being negative.

50. In a class of 40 students, 25 study French, 18 study German and 9 study both languages. How many students study neither language?

- A)** 6 **B)** 9 **C)** 15 **D)** 16 **E)** 34

Answer: A

Given: class of 40; 25 study French, 18 study German, 9 study both.

Relationship: inclusion-exclusion, students taking at least one language = French + German – both, because the 9 taking both were counted twice.

Substitute: $25 + 18 - 9$.

Step: $25 + 18 = 43$

Step: $43 - 9 = 34$ study at least one language.

Step: $40 - 34 = 6$ study neither

Result: 6 students study neither language.

The tempting 15 counts only the students outside French and forgets the German-only students, and 34 stops at the union without subtracting it from the class size.

51. If $3x + 2y = 12$ and $y = 3$, then x equals:

- A) 1 B) 2 C) 3 D) 4 E) 6**

Answer: B

Given: $3x + 2y = 12$ and $y = 3$

Step 1: substitute $y = 3$, giving $3x + 2 \times 3 = 3x + 6 = 12$

Step 2: subtract 6 from both sides, giving $3x = 12 - 6 = 6$.

Step 3: divide both sides by 3, giving $x = 6 \div 3 = 2$.

Result: $x = 2$

The tempting 4 comes from dividing 12 by 3 before removing the 6, and copying the given y supplies the spurious 3.

52. The expression $x^2 - 16$ can be factorised as:

- A) $(x - 4)^2$ B) $(x + 4)^2$ C) $x(x - 16)$ D) $(x - 2)(x + 8)$ E) $(x - 4)(x + 4)$**

Answer: E

Given: the expression $x^2 - 16$.

Relationship: difference of squares, $a^2 - b^2 = (a - b)(a + b)$

Substitute: $16 = 4^2$, so $x^2 - 16 = x^2 - 4^2 = (x - 4)(x + 4)$

Check by expanding: $(x - 4)(x + 4) = x^2 + 4x - 4x - 16 = x^2 - 16$, the middle terms cancel.

Result: $x^2 - 16 = (x - 4)(x + 4)$

The tempting $(x - 4)^2$ expands to $x^2 - 8x + 16$, carrying a linear term the original lacks, and $x(x - 16)$ expands to $x^2 - 16x$.

53. A right-angled triangle has legs of 6 cm and 8 cm. Its perimeter is:

- A) 14 cm B) 20 cm C) 24 cm D) 26 cm E) 30 cm**

Answer: C

Given: right-angled triangle with legs 6 cm and 8 cm.

Relationship: Pythagoras' theorem, hypotenuse = $\sqrt{(\text{first leg}^2 + \text{second leg}^2)}$.

Substitute: hypotenuse = $\sqrt{6^2 + 8^2}$ cm

Step: $6^2 = 36$ and $8^2 = 64$, so the sum is $36 + 64 = 100 \text{ cm}^2$

Step: $\sqrt{100 \text{ cm}^2} = 10 \text{ cm}$, so the hypotenuse is 10 cm

Step: perimeter = 6 cm + 8 cm + 10 cm = 24 cm

Result: the perimeter is 24 cm.

The tempting 14 cm adds only the two legs and leaves out the hypotenuse entirely.

54. In a right-angled triangle, one angle is 30° and the hypotenuse has length 10 cm. The side opposite the 30° angle has length:

- A)** 2.5 cm **B)** 5 cm **C)** $5\sqrt{3}$ cm **D)** 10 cm **E)** 12 cm

Answer: B

Given: right-angled triangle, one angle 30° , hypotenuse 10 cm.

Relationship: sine of an angle = opposite side $\div$ hypotenuse, so opposite = hypotenuse $\times$ sin(angle).

Substitute: opposite = 10 cm $\times$ sin(30°)

Step: sin(30°) = $1/2$ exactly

Step: 10 cm $\times$ $1/2$ = 5 cm

Result: the side opposite the 30° angle is 5 cm.

The tempting $5\sqrt{3}$ cm is the side adjacent to the 30° angle, obtained by multiplying by cos(30°) = $\sqrt{3}/2$ instead of the sine.

Section E - Physics (Q55-60)

55. The velocity-time graph of a cyclist is a horizontal line at 6 m/s for the whole interval from $t = 0$ to $t = 10$ s. The distance covered in this interval is:

- A)** 0.6 m **B)** 6 m **C)** 30 m **D)** 60 m **E)** 600 m

Answer: D

Given: constant velocity $v = 6$ m/s from $t = 0$ s to $t = 10$ s

Relationship: at constant velocity, distance = velocity $\times$ time, which is the area of the rectangle under the velocity-time graph.

Substitute: distance = 6 m/s $\times$ 10 s

Step: $6 \times 10 = 60$, so distance = 60 m

Result: the cyclist covers 60 m.

The tempting 30 m halves the speed as if the cyclist accelerated uniformly from rest, and 600 m multiplies by 100 instead of 10.

56. When a 12 V battery is connected across a lamp, a current of 0.5 A flows. The resistance of the lamp is:

- A)** 6 Ω **B)** 12 Ω **C)** 24 Ω **D)** 48 Ω **E)** 60 Ω

Answer: C

Given: $V = 12$ V across the lamp, $I = 0.5$ A through it

Relationship: Ohm's law, resistance $R = V \div I$.

Substitute: $R = 12$ V $\div$ 0.5 A

Step: dividing by 0.5 doubles the value, so $R = 12 \times 2 = 24$ Ω

Result: the resistance of the lamp is 24 Ω .

The tempting 6 Ω multiplies instead of dividing, $12 \times 0.5 = 6$, which is in fact the power in watts, and 48 Ω doubles the voltage a second time.

57. A 60 W lamp is left switched on for 2 hours. The electrical energy it consumes is:

- A)** 0.012 kWh **B)** 0.12 kWh **C)** 1.2 kWh **D)** 12 kWh **E)** 120 kWh

Answer: B

Given: $P = 60 \text{ W}$ left on for $t = 2 \text{ hours}$

Relationship: energy = power $\times$ time; in kilowatt-hours the power must be in kW and the time in hours.

Step: $60 \text{ W} = 0.06 \text{ kW}$

Substitute: $E = 0.06 \text{ kW} \times 2 \text{ h}$

Step: $0.06 \times 2 = 0.12$, so $E = 0.12 \text{ kWh}$

Result: the lamp consumes 0.12 kWh.

Every tempting neighbouring value merely shifts the decimal point, for example 1.2 kWh from misreading 60 W as 0.6 kW.

58. A box slides across a level floor at constant velocity. The resultant (net) force acting on the box is:

- A)** directed forwards and larger than the friction force.
B) equal to the friction force and directed forwards.
C) equal to the weight of the box.
D) directed backwards.
E) zero.

Answer: E

Restate: the box slides across a level floor at constant velocity; find the resultant force.

Deduction 1: constant velocity means zero acceleration.

Deduction 2: Newton's second law gives resultant force = mass $\times$ acceleration = mass $\times 0 = 0$.

Deduction 3: the individual forces do not vanish; the forward push exactly cancels the backward friction, and the weight is separately cancelled by the normal reaction from the floor.

Conclusion: the resultant force on the box is zero.

The tempting force equal to friction but directed forwards is the push itself, a single force rather than the resultant; any non-zero resultant would accelerate the box and change its velocity.

59. An astronaut weighs about one sixth as much standing on the Moon as standing on Earth because:

- A)** the Moon's smaller gravitational acceleration acts on the astronaut's unchanged mass.
B) the astronaut's body contains less matter on the Moon.
C) the Moon has no atmosphere to push up on the astronaut.
D) weight cannot be measured where there is no water or air.
E) part of the astronaut's weight is still carried by the distant Earth.

Answer: A

Given: mass m of the astronaut, unchanged by location; $g(\text{Earth}) \approx 9.8 \text{ m/s}^2$; $g(\text{Moon}) \approx 1.6 \text{ m/s}^2$.

Relationship: weight $W = \text{mass} \times \text{local gravitational acceleration}$.

Step: ratio of weights = $1.6 \div 9.8 \approx 0.16 \approx 1/6$

Step: $9.8 \div 6 \approx 1.6 \text{ m/s}^2$, confirming that the Moon's smaller mass and radius give about one sixth of Earth's surface gravity.

Result: the same mass multiplied by a six times smaller g , so the astronaut weighs one sixth as much on the Moon.

The tempting claim that the astronaut contains less matter confuses weight with mass, which is identical everywhere, and the absence of an atmosphere plays no part in defining weight.

60. An object weighs 50 N in air. When it is fully submerged in water, its apparent weight is 40 N. The buoyant force exerted by the water on the object is:

- A)** 5 N **B)** 10 N **C)** 40 N **D)** 50 N **E)** 90 N

Answer: B

Given: true weight in air = 50 N; apparent weight fully submerged = 40 N.

Relationship: the water pushes up with a buoyant force, so apparent weight = true weight – buoyant force, hence buoyant force = true weight – apparent weight.

Substitute: buoyant force = 50 N – 40 N

Step: $50 - 40 = 10$, so buoyant force = 10 N

Result: the buoyant force is 10 N, which by Archimedes' principle equals the weight of the displaced water.

The tempting 40 N is the apparent weight itself rather than the upward force, and 90 N adds the two weights where the physics subtracts them.